Smart Campus Hostel Management

Story

A university wants to digitize hostel operations. Students are assigned to rooms across multiple hostels. Each hostel has a warden. Students can raise maintenance complaints, which are handled by maintenance staff.

Technical Requirements

- Store student details (ID, name, contact info)
- Hostels contain multiple rooms
- Each room has a fixed capacity
- Students are assigned to rooms
- Each hostel has one or more wardens
- Students can raise multiple complaints
- Complaints are assigned to maintenance staff
- Track complaint status and timestamps

What Students Should Do

- Design tables and relationships
- Decide appropriate primary keys and foreign keys
- Handle one-to-many and many-to-many relationships
- Normalize schema up to 3NF
- Avoid redundant storage (e.g., repeating hostel or room data in student table)

Evaluation Focus

- Room allocation modeling
- Complaint assignment structure
- Handling multi-valued attributes like phone numbers
- Avoiding transitive dependencies (e.g., hostel → warden stored incorrectly in student)

Mini E-Commerce Platform

Story

A startup is building an online shopping platform. Customers can browse products, place orders, and make payments. Each order can include multiple products, and products belong to categories.

Technical Requirements

- Store customer details
- Products belong to categories
- Customers place orders
- Each order can contain multiple products
- Track quantity and price at the time of purchase
- Support multiple payment methods per order
- Orders have statuses (placed, shipped, delivered)
- Optional discount coupons

What Students Should Do

- Identify and design junction tables
- Separate product price history from current price
- Design order and order-item relationship properly
- Normalize up to 3NF
- Avoid duplication of product or customer data in orders

Evaluation Focus

- Correct handling of many-to-many (orders ↔ products)
- Use of composite vs surrogate keys
- Payment modeling (one-to-many vs one-to-one)
- Avoiding anomalies in price and order data

Online Learning Platform

Story

An edtech platform allows instructors to create courses. Students can enroll in courses and watch lessons. Each course contains multiple modules and lessons. Students' progress is tracked.

Technical Requirements

- Store student and instructor details
- Instructors create multiple courses
- Courses contain modules, which contain lessons
- Students enroll in courses
- Track student progress per lesson
- Students can rate and review courses
- Courses can have multiple tags (e.g., "Web Dev", "AI")

What Students Should Do

- Design hierarchical relationships (course → module → lesson)
- Model enrollments properly
- Track progress without redundancy
- Handle many-to-many relationships (courses ↔ tags, students ↔ courses)
- Normalize up to 3NF

Evaluation Focus

- Proper decomposition of course structure
- Avoiding storing aggregated progress redundantly
- Designing review and rating system
- Handling multi-valued attributes (tags) correctly